ASSIGNMENT SOLUTIONS: Cryptography

COURSE	SUBJECT CODE	SUBJECT TITLE	SUBMISSION DATE
BCA	BCA309T-AS01	Foundations of Cryptography	05 October 2026

Q1. Define cryptography in your own words. What are the three main goals it tries to achieve? SHORT ANSWER

Answer: Cryptography is the science and mathematical art of securing communication and data so that only intended recipients can access and understand it. It transforms plain readable information into an unreadable format to safeguard it from unauthorized interception or tampering across untrusted networks.

The three main goals of cryptography (collectively known as the **CIA Triad**) are:

- **Confidentiality:** Ensuring that sensitive information remains completely private and inaccessible to unauthorized entities.
- **Integrity:** Guaranteeing that data has not been altered, modified, or corrupted during transmission or storage.
- **Availability:** Ensuring that authorized users have timely and reliable access to systems and data whenever required.

Q2. Differentiate between plaintext and ciphertext with a simple example. SHORT ANSWER

Answer: Plaintext refers to original, unencrypted data or human-readable text that anyone can read directly. **Ciphertext** is the scrambled, unintelligible output generated after applying an encryption algorithm and key to the plaintext.

Simple Example:
• *Plaintext:* `"MEET AT 5 PM"` (Readable and understandable)
• *Encryption Rule:* Shift each letter forward by 1 position (Caesar Cipher key = +1)
• *Ciphertext:* `"NFFU BU 5 QN"` (Scrambled text that reveals no meaning to eavesdroppers)

Q3. Explain the concept of cryptography in detail. Discuss why it is essential in the modern digital world, giving at least three real-life applications. LONG ANSWER

Answer:

Concept of Cryptography: At its core, cryptography relies on mathematical algorithms (ciphers) and key management to protect data in transit (e.g., streaming or downloading) and data at rest (e.g., hard drives or cloud servers). It moves beyond basic secrecy to establish trust, identity verification, and non-repudiation in digital interactions.

Essential Need in the Modern Digital World: Modern society relies almost entirely on digital infrastructure—from online banking to government utilities. Because open communication channels like the Internet are inherently insecure and prone to eavesdropping, packet sniffing, and cyberattacks, cryptography serves as the primary security backbone that safeguards privacy and enables safe global digital commerce.

Real-Life Applications:

1. **Online Banking & Financial Transactions:** Financial platforms use SSL/TLS protocols and end-to-end encryption to protect card numbers, PINs, and account credentials during transfers, preventing financial fraud.

Q4. What is the difference between encryption and decryption? SHORT ANSWER

Answer: Encryption is the forward cryptographic process of converting original, readable data (Plaintext) into a coded format (Ciphertext) using a secret key or mathematical algorithm at the sender's end. Conversely, **Decryption** is the reverse process of converting coded data (Ciphertext) back into its original readable format (Plaintext) using an appropriate decryption key at the receiver's end.

Q5. Describe Symmetric and Asymmetric Key Cryptography. Compare them based on key usage, speed, security, and use cases with algorithm examples. LONG ANSWER

Answer:

1. Symmetric Key Cryptography: Uses a *single shared secret key* for both encryption and decryption. Both sender and receiver must securely exchange and possess this secret key beforehand. It is exceptionally fast and efficient for processing large volumes of data.

2. Asymmetric Key Cryptography (Public-Key Cryptography): Uses a mathematically linked *key pair*: a **Public Key** (shared openly for encryption) and a **Private Key** (kept strictly secret by the owner for decryption). It solves the complex key distribution problem inherent in symmetric systems.

Detailed Comparison Matrix:

Feature	Symmetric Key Cryptography	Asymmetric Key Cryptography
Key Usage	Single secret key shared by both parties.	Pair of keys: Public key (encrypt), Private key (decrypt).
Speed & Performance	Very fast with minimal computational overhead.	Slower due to complex mathematical computations.
Security & Key Exchange	High security, but secure key exchange is difficult.	High security; solves key distribution efficiently.
Data Size	Ideal for bulk data encryption.	Best for small data (e.g., keys, digital signatures).
Typical Use Cases	Disk encryption (BitLocker), SSL/TLS payload encryption.	Digital signatures, SSL/TLS handshake, SSH keys.
Algorithms	**AES** (Advanced Encryption Standard), **DES**, 3DES.	**RSA**, **ECC** (Elliptic Curve Cryptography), Diffie-Hellman.

Q6. What is hashing, and how is it different from encryption? SHORT ANSWER

Answer: Hashing is a one-way mathematical function that converts arbitrary-length input data into a fixed-length string of characters (a hash value or digest). Unlike encryption, hashing is *irreversible*—there is no key used to turn a hash back into the original plaintext. Encryption is a two-way function intended for confidentiality and message recovery, whereas hashing is designed to verify data integrity (e.g., verifying file downloads or storing password hashes safely).

Q7. Critically evaluate why achieving confidentiality, integrity, and non-repudiation together is challenging in real-world communication systems. SHORT ANSWER

Answer: Achieving confidentiality, integrity, and non-repudiation simultaneously is challenging because each goal requires integrating different cryptographic primitives (e.g., symmetric ciphers, hash functions, asymmetric digital signatures, and Public Key Infrastructure). Combining these mechanisms introduces significant computational latency, increased bandwidth costs, and complex key management overhead. Additionally, human errors—such as compromised private keys or poorly configured server certificates—frequently undermine the overall security model despite mathematical soundness.

Q8. Explain the role of cryptography in network security. Discuss how it helps achieve confidentiality, integrity, authentication, and non-repudiation, with examples. LONG ANSWER

Answer: Cryptography forms the bedrock of network security by transforming unsecured, open channels into trusted communication pathways. It prevents eavesdropping, tampering, impersonation, and false denials across computer networks.

- **1. Confidentiality:** Ensures that intercepted network traffic cannot be read by packet sniffers or unauthorized third parties.
  Example: Using **AES-256 encryption** in HTTPS protocols to protect online credit card payments from Wi-Fi eavesdroppers.
- **2. Integrity:** Ensures that data sent across network routers arrives at its destination without unauthorized modification or corruption.
  Example: Appending a **HMAC (Hash-based Message Authentication Code)** or SHA-256 hash digest to network packets to verify they were not altered in transit.
- **3. Authentication:** Verifies the true identity of connecting entities (users, servers, or devices) before granting access to resources.
  Example: Using **X.509 SSL Certificates** so a web browser can confirm it is connected to the real bank server rather than a phishing site.
- **4. Non-Repudiation:** Prevents a sender from falsely claiming they never transmitted a specific message or initiated a transaction.
  Example: Signing legal contracts or high-value wire transfers using **RSA Digital Signatures**, where only the sender's unique private key could have generated the signature.

Q9. Name any two network security protocols that use cryptographic techniques and briefly state their purpose. SHORT ANSWER

Answer:

1. **SSL/TLS (Secure Sockets Layer / Transport Layer Security):** Encrypts communications between web browsers and servers, providing confidentiality, integrity, and authentication for HTTPS websites and API endpoints.
2. **IPSec (Internet Protocol Security):** Operates at the IP network layer to authenticate and encrypt all data packets passed between two remote systems, commonly used to build secure Virtual Private Networks (VPNs).

Q10. Critically discuss the evolution of cryptography from classical ciphers to modern systems, evaluating the impact of computational advances. LONG ANSWER

Answer:

Cryptography evolved from basic physical or linguistic pencil-and-paper ciphers to mathematically complex computer algorithms. Classical ciphers (e.g., substitution and transposition techniques) relied heavily on security through obscurity or key secrecy, but were rendered insecure by basic frequency analysis and letter pattern identification.

Impact of Computational Advances: As computer processing power exploded according to Moore's Law, classical and early mechanical ciphers became trivial to break via brute-force automated testing. This forced modern cryptography to shift toward mathematically hard problems—such as prime factorization (RSA) and discrete logarithms (ECC). Today, standard algorithms like AES-256 are computationally unbreakable by classical hardware, though the upcoming rise of quantum computing is driving the next evolution toward Post-Quantum Cryptography (PQC).

Q11. Trace the historical development of cryptographic techniques from classical methods to modern computer-based cryptography. LONG ANSWER

Answer: The historical development of cryptography spans four key evolutionary eras:

- **1. Classical Era (Ancient Times – 19th Century):** Early systems relied on manual character shifting. The *Caesar Cipher* used simple monoalphabetic substitution, while the *Vigenère Cipher* introduced polyalphabetic substitution using a keyword, standing uncracked for centuries until frequency analysis techniques improved.
- **2. Mechanical Era & WWII (Mid 20th Century):** The invention of electromechanical rotor machines revolutionized secrecy. Germany's **Enigma Machine** generated continuously shifting substitution ciphers. Alan Turing and the cryptanalysts at Bletchley Park developed the electro-mechanical *Bombe* machine, marking a major turning point that birthed modern computer science.
- **3. Early Computer Era & Symmetric Standards (1970s):** Digital computers enabled bit-level operations. IBM and NIST standardized the **DES (Data Encryption Standard)** in 1977. When 56-bit DES keys became vulnerable to brute-force attacks in the late 1990s, NIST replaced it with the modern **AES (Advanced Encryption Standard)**.
- **4. Public-Key Revolution (1976 – Present):** The shift to distributed computer networks created a massive key-sharing problem, leading to the invention of asymmetric public-key cryptography (RSA and Elliptic Curve Cryptography), enabling secure global commerce today.

Q12. Who introduced the concept of public-key cryptography, and in which year? SHORT ANSWER

Answer: The concept of public-key cryptography was publicly introduced by **Whitfield Diffie** and **Martin Hellman** in their landmark research paper titled *"New Directions in Cryptography,"* published in **1976**. (Note: Ralph Merkle also contributed foundational ideas with Merkle's Puzzles during the same era).